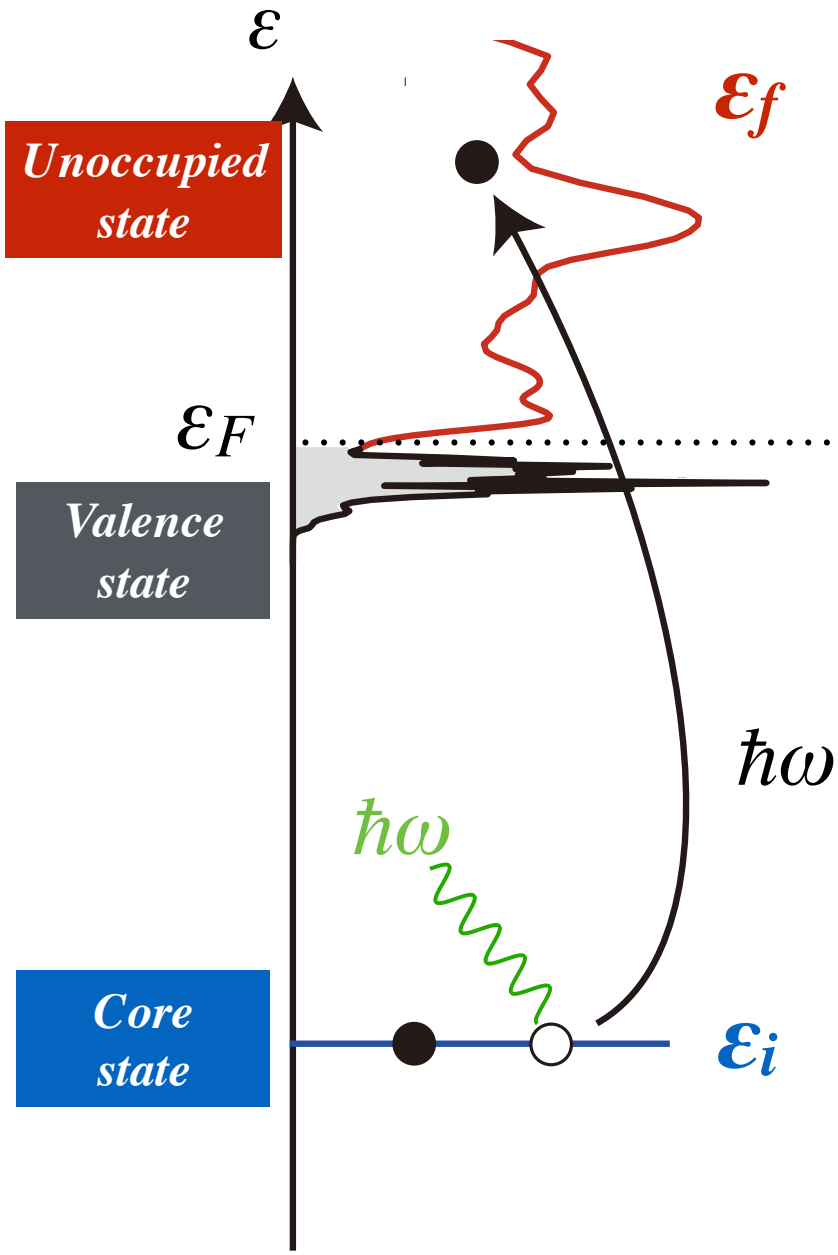
XANES スペクトルの起源(1)

遷移確率　フェルミの黄金律

$$W(\omega) = \frac{2\pi}{\hbar}\sum_f \left|\langle f|\hat{F}|i\rangle\right|^2 \delta(\varepsilon_f - \varepsilon_i - \hbar\omega)$$

unoccupied state　core state

終状態　始状態



Cu-p (LDOS) with hole

遷移確率

$$W(\omega)=\frac{2\pi}{\hbar}\sum_f\left|\langle f|\hat{F}|i\rangle\right|^2\delta(\varepsilon_f-\varepsilon_i-\hbar\omega)$$

始状態 $|i\rangle=\sum_{mm_s}|nlmm_s\rangle c^{(nl)}_{mm_s,i}$ $\qquad \phi_{nlmm_s}=R_{nlm_s}Y_{lm}\sigma_{m_s}$

終状態 $|f\rangle=\sum_{LMM_s}|LMM_s\rangle a_{LMm_s,f}$ $\qquad \phi_{LMM_s}=R_{LM_s}Y_{LM}\sigma_{M_s}$

電気双極子 $\hat{F}=e\mathrm{E}\cdot r=e(E_x x+E_y y+E_z z)$

…

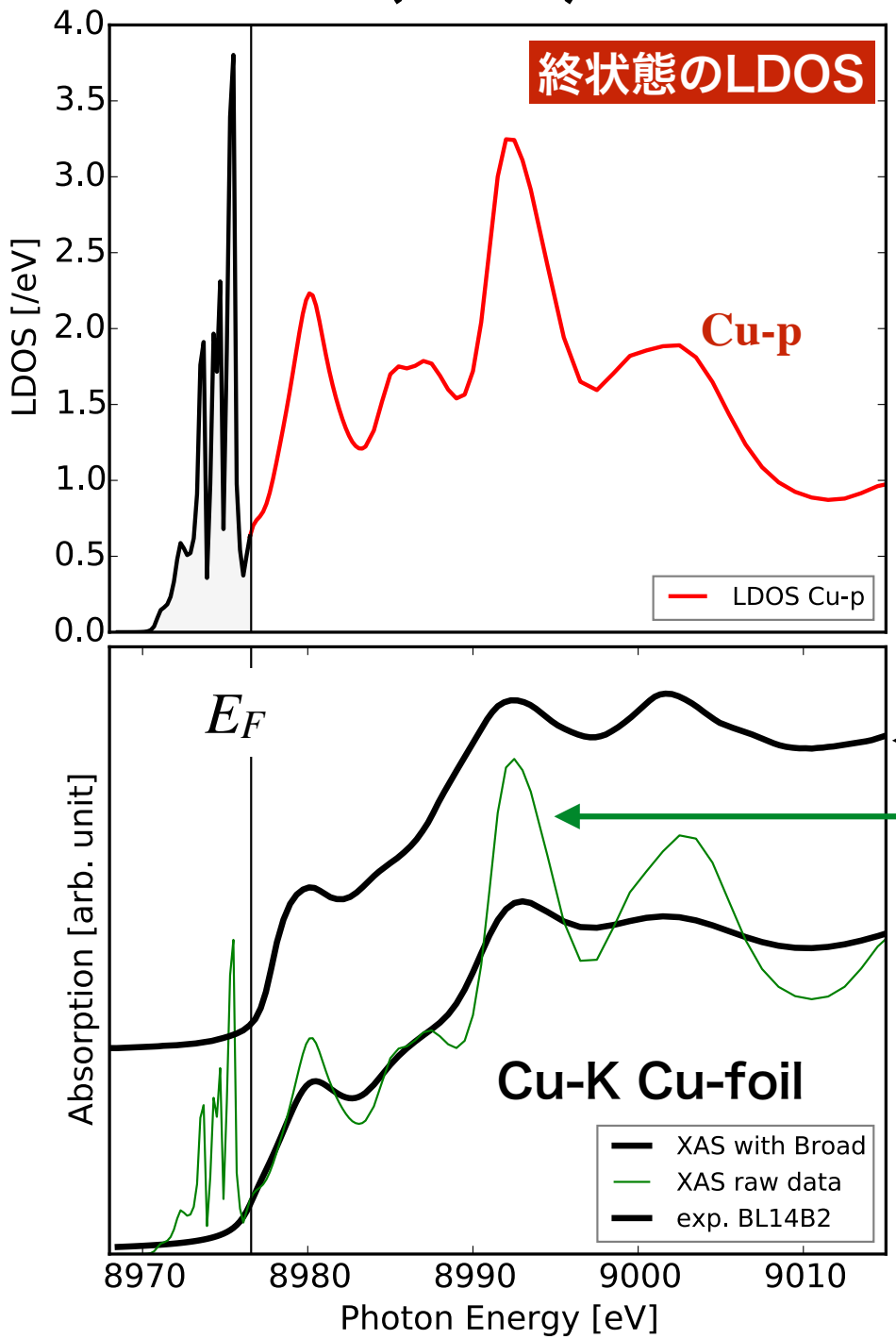
$$\propto\left|I_{Lnl}\right|^2\sum_{m_s}w_{j\mu m_s}\boxed{D_{LMm_s}(\hbar\omega+\varepsilon_i)}$$

終状態の部分状態密度

部分状態密度 $D_{LMM_s}(\varepsilon)=\sum_f\left|a_{LMM_s,f}\right|^2\delta(\varepsilon_f-\varepsilon)$

動径積分 $I^{(k)}_{LM_s,nlm_s}(\varepsilon_f)=\int_0^\infty R_{LM_s}(r;\varepsilon_f)r^kR_{nlm_s(r)}\cdot r^2dr$

FDMNES (R=7.0) FDM



終状態のLDOS

$$W^{(h)}(\omega) \propto \left| I_{Lnl} \right|^2 \sum_{m_s} w^{(h)}_{j\mu m_s} D_{LMm_s}(\hbar\omega + \varepsilon_i)$$

$$\begin{cases} L = l \pm 1 \\ M = m + h \\ M_s = m_s \end{cases}$$

実験値

計算値(遷移確率)

計算値 (Lorentzian Broadening)

XANES スペクトル≒局所状態密度

始状態 $$\left|i\right\rangle = \sum_{mm_s} \left|nlmm_s\right\rangle c^{(nl)}_{mm_s,i}$$ $$\phi_{nlmm_s} = R_{nlm_s} Y_{lm} \sigma_{m_s}$$

終状態 $$\left|f\right\rangle = \sum_{LMM_s} \left|LMM_s\right\rangle a_{LMm_s,f}$$ $$\phi_{LMM_s} = R_{LM_s} Y_{LM} \sigma_{M_s}$$

$$\left\langle f\right|\hat{F}\left|i\right\rangle = \sum_{LMM_s} a^*_{LMm_s,f} c^{(nl)}_{mm_s,i} F^{(h)}_{LMM_s,nlmm_s}$$

$$F^{(h)}_{LMM_s,nlmm_s} = e \cdot I^{(1)}_{LM_s,nlm_s} \cdot g(LM,1h,lm) \cdot \delta_{M_s m_s}$$ I: 動径積分

$$D_{LMM_s}(\varepsilon) = \sum_f \left|a_{LMM_s,f}\right|^2 \delta(\varepsilon_f - \varepsilon)$$

$$W(\omega) = \frac{2\pi}{\hbar} \sum_f \left|\left\langle f\right|\hat{F}\left|i\right\rangle\right|^2 \delta(\varepsilon_i + \hbar\omega - \varepsilon_f)$$

$$I^{(k)}_{LM_s,nlm_s}(\varepsilon_f) = \int_0^\infty R_{LM_s}(r;\varepsilon_f) r^k R_{nlm_s(r)} \cdot r^2 dr$$

遷移確率

動径積分・Gaunt係数

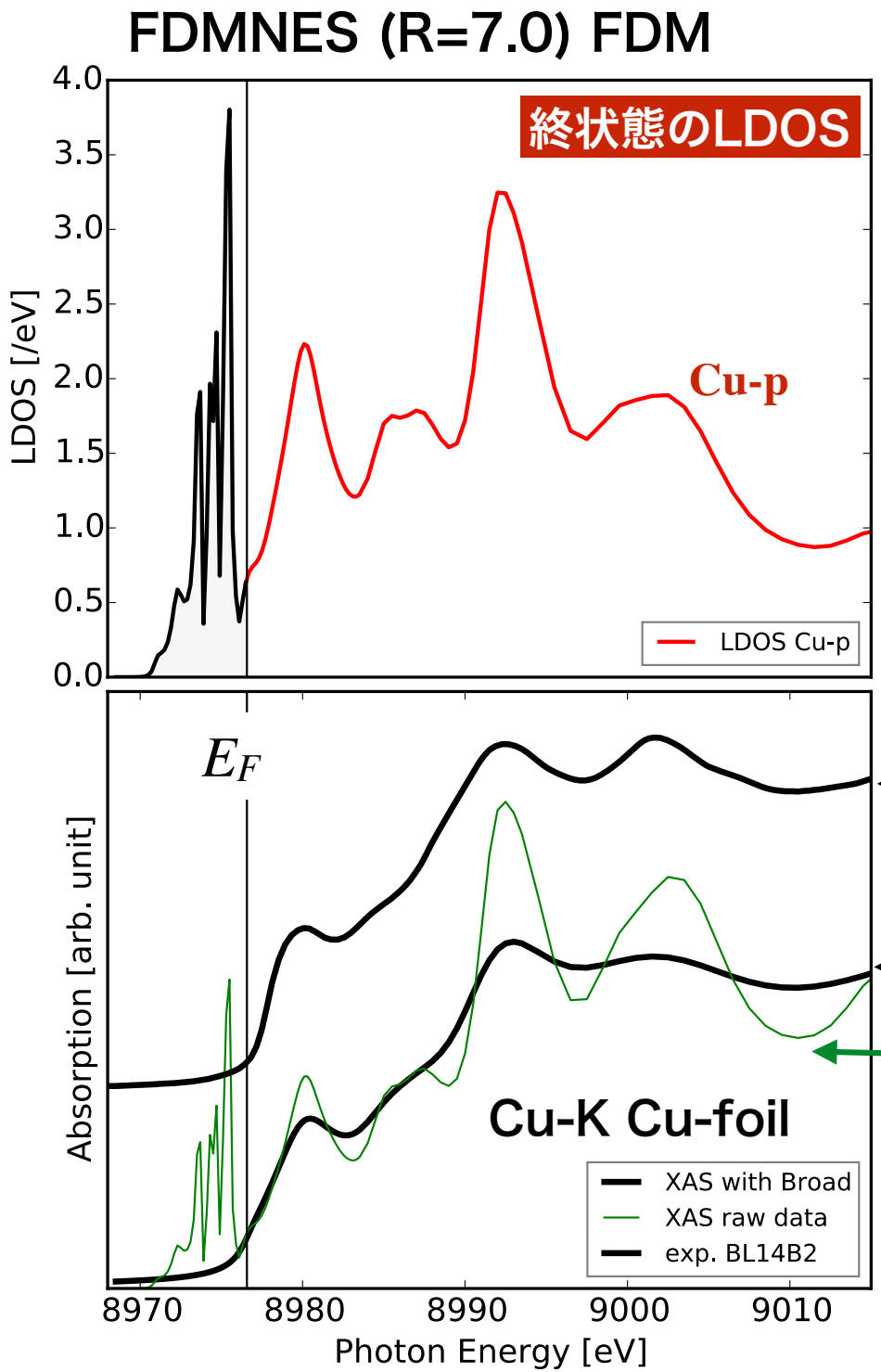
$$\left|\langle f|\hat{F}|i\rangle\right|^2 = \sum_{LMM_s mm_s} a^*_{LMm_s,f}\, c^{(nl)}_{mm_s,i} \cdot F_{LMM_s,nlmm_s}$$

終状態のDOS　始状態のDOS

式を見直す

$$W(\omega) \propto \left|I_{Lnl}\right|^2 \sum_{m_s} w_{j\mu m_s} D_{LMm_s}(\hbar\omega + \varepsilon_i)$$

終状態の状態密度



実験値

計算値 (Lorentzian Broadening)

計算値

XANES スペクトル≒局所状態密度